

Censoring and truncation in astronomical surveys

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**Statistical Modeling of Cosmic Populations
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Selection bias in flux-limited surveys

Astronomers often struggle to detect faint characteristics of celestial populations and fail. Many surveys are flux-limited ($F=L/4\pi d^2$), and are limited to detecting the closer and/or more luminous members of a population. This leads to biased samples: at large distances, high-luminosity objects are over-represented (e.g., the majority of $V<2$ stars are giants and supergiants). Two types of bias in flux-limited surveys:

1. A 'blind' astronomical survey of a portion of the sky is thus truncated at the sensitivity limit, where **truncation** indicates that the undetected objects, even the number of undetected objects, are entirely missing from the dataset.
2. In a 'supervised' astronomical survey where a particular property (e.g., IR emission with Herschel, HCO^+ line emission with ALMA, redshifted $\text{Ly}\alpha$ emission with HETDEX) of a previously defined sample of objects is sought, some objects in the sample may be too faint to detect. This gives **upper limits** or **left-censored** datapoints.

Statistical challenges of censoring & truncation

In a truncated or censored sample, neither the same mean nor the median converge to the population values.

The sample distribution (e.g. a Pareto luminosity function $N(L) \sim L^{-\alpha}$) will not converge to the population distribution because faint objects are underrepresented.

Relationships among variables (e.g. $L_{\text{opt}} \sim L_{\text{radio}}^{-\alpha}$) may be less affected, but sample correlations will be biased unless nondetections are adequately treated.

Survival analysis

A large field of applied statistics called **survival analysis** developed during 1950-80s to treat right-censoring in several applications:

1. **Life insurance** To calculate annuities, Edmund Halley (1692) constructed 'life tables' from birth/death records in a city. But some people leave the city; this leads to right-censoring in their survival time. (Need: Univariate distribution function)
2. **Industrial reliability** A company manufactures Widget Mark IV. To find improvement over Mark III, operate 100 widgets until 20% fail. From failure times, compare lifetime distributions. (Need: 2-sample test)
3. **Biometrics** Dangerous Tobacco Co. wants to test the effect of smoking on cancer rates. To compare longevities, 100 rats are given smoke at 0-5 cigarette packs/day. After 1 year, the experiment is stopped but some rats are still alive with right-censored survivals. (Need: Regression)
4. **Astronomy** The VLA seeks 21cm hydrogen line emission from starburst galaxies. Due to low star formation rate and/or large distance, half are not detected. Compare to LINERs and infrared dust emission. (Need: distribution function, 2-sample test, regression)

Concepts of survival analysis

Censoring: The sample is known and all objects are observed, but some are undetected in the desired property. They can be displayed on graphs with 'arrows' rather than 'points'. Synonyms: Upper limits = Nondetections = Left censoring. Lower limits = right censoring.

Truncation: An unknown number of objects are missing from the sample due to nondetection. They cannot be displayed on graphs.

Survival and hazard functions: Univariate functions closely related to the e.d.f. and p.d.f. widely used in survival analysis.

Proportional hazard model and Cox regression: A mathematically convenient form of the dependence of the hazard function on uncensored variables.

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Statistical foundations of survival analysis

The **survival function** $S(x)$ gives the probability that an object has a value of X above a specified value x (the inverse of the e.d.f.):

$$\begin{aligned} S(x) &= \text{Prob}(X > x) = \frac{\# \text{observation} \geq x}{n} \\ &= 1 - F(x) = 1 - \int_0^x f(s) ds. \end{aligned}$$

The **hazard rate** gives the probability that an object will have a specified value x :

$$h(x) = \frac{f(x)}{S(x)} = -\frac{d \ln S(x)}{dx}$$

(Note the p.d.f. is the product of the survival function and hazard rate.)

Example: Pareto distribution

$$f(x) = \frac{\theta \lambda^\theta}{x^{\theta+1}}$$

$$\begin{aligned} S(x) &= \frac{\lambda^\theta}{x^\theta} \\ h(x) &= \frac{\theta}{x}. \end{aligned}$$

Kaplan-Meier estimator

For a randomly censored univariate X , the Kaplan-Meier estimator is the unique unbiased nonparametric maximum likelihood estimator of the survival function is

$$\hat{S}_{KM}(x) = \prod_{x_i \geq x} \left(1 - \frac{d_i}{N_i} \right)$$

where d_i is the number of occurrences at x_i ($d_i=1$ if no ties are present) and N_i is the number of 'at risk' objects left in the sample. An intuitive procedure: construct the e.d.f. of detected points, but increase the step size at low values by redistributing the upper limits to the left.

The Kaplan-Meier estimator is asymptotically normal with variance

$$\widehat{Var}(\hat{S}_{KM}) = \hat{S}_{KM}^2 \sum_{x_i \geq x} \frac{d_i}{N_i(N_i - d_i)}$$

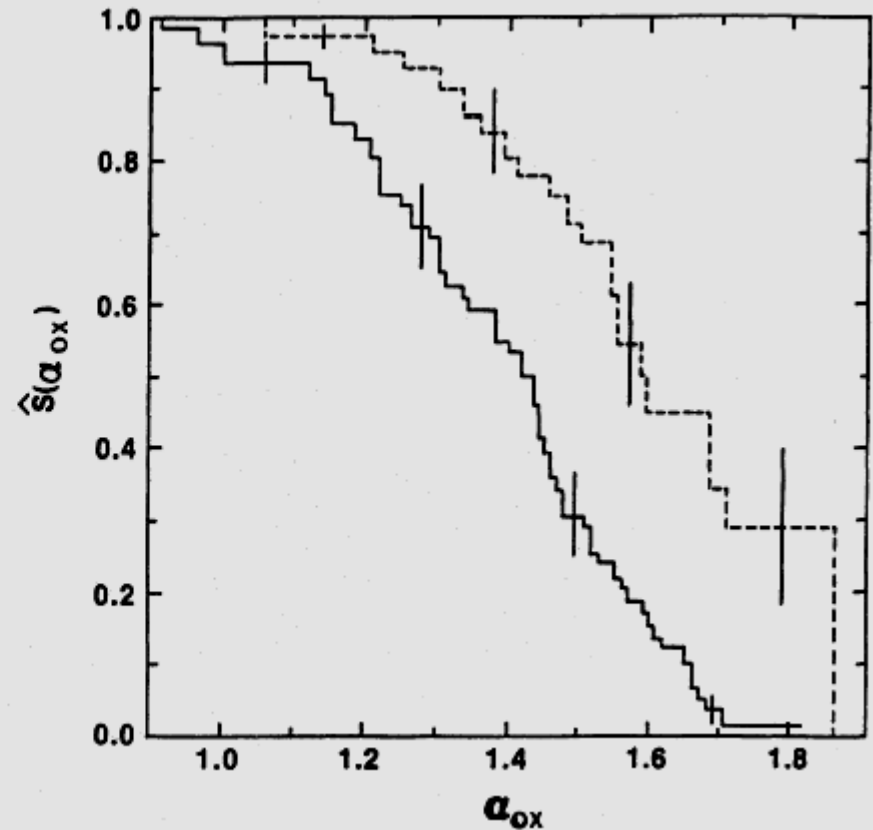
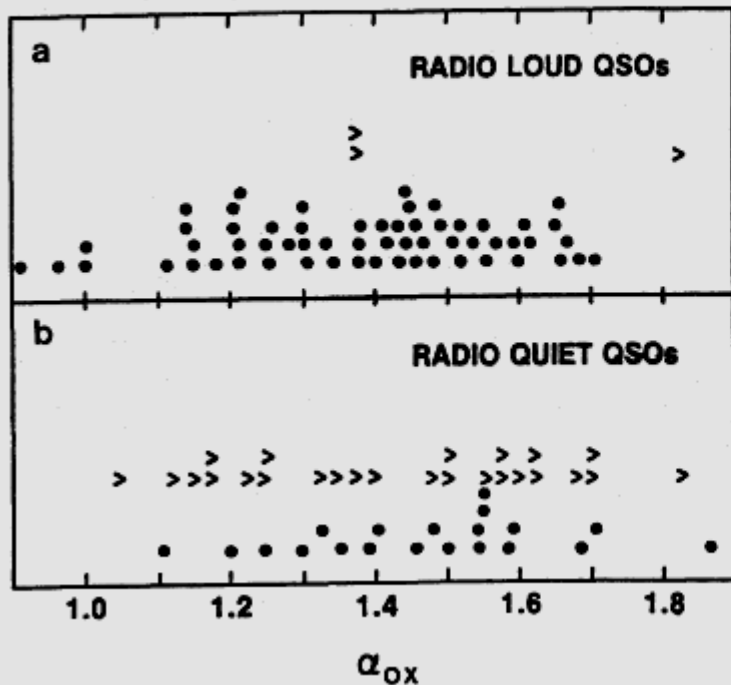
Kaplan, E. L.; Meier, P.: Nonparametric estimation from incomplete observations.

J. Amer. Statist. Assn. 53:457–481, 1958.

Greenwood, M. The natural duration of cancer. Rpt Public Health (London) 33:1, 1926

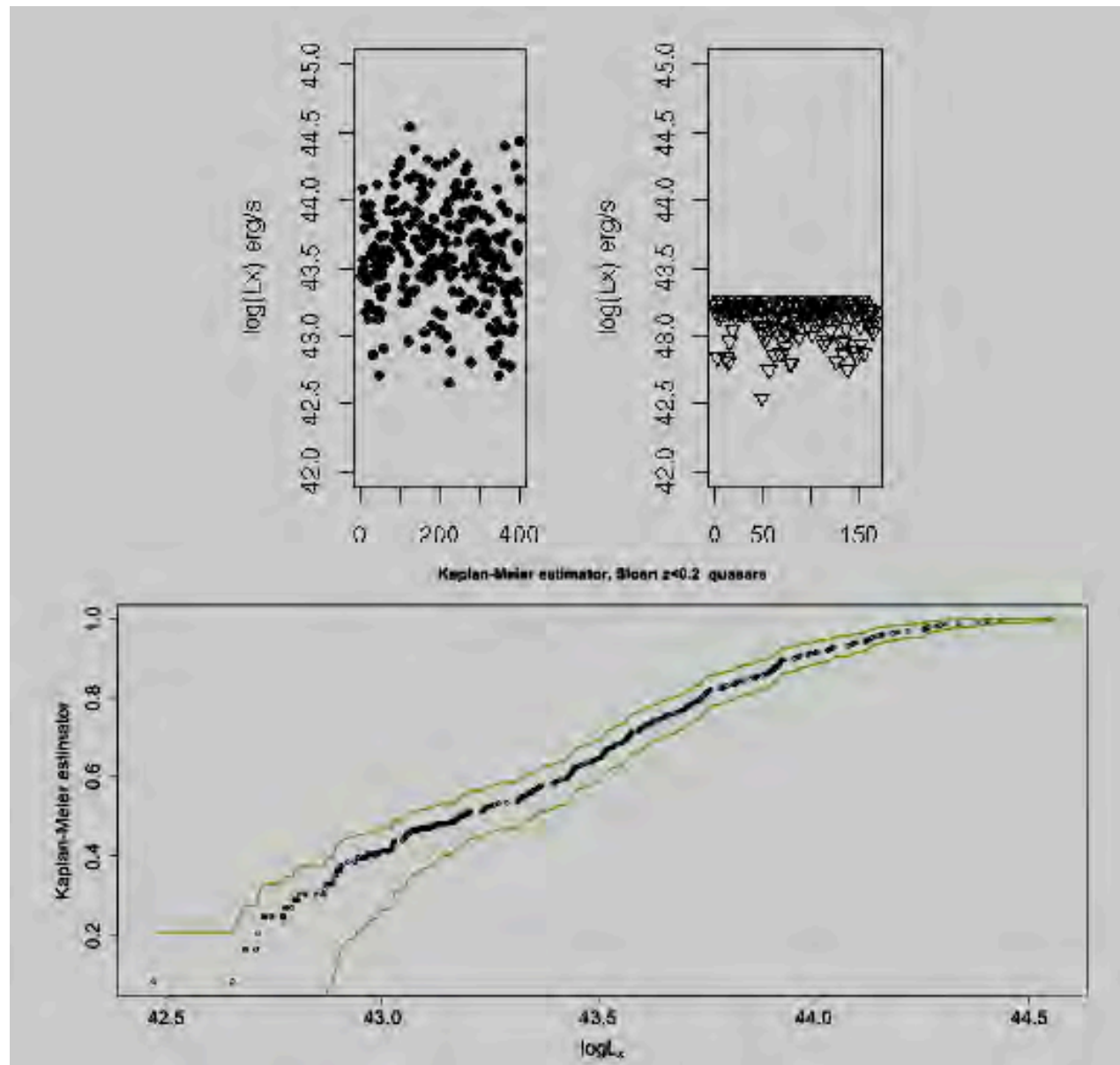
Example of univariate survival analysis (Kaplan-Meier)

Redistribute-to-the-right algorithm (Efron)



Feigelson & Nelson 1986

Example:
X-ray
luminosity
function of
quasars from
ROSAT All-Sky
Survey



R script:

```
survobj <- Surv(-XLum, Xstatus)
KM.XLF<- survfit(survobj~1, conf.int=0.95,
```

```
plot(log10(-KM.XLF$time), log10(KM.XLF$surv),
lines(log10(-KM.XLF$time), KM.XLF$upper,
lines(log10(-KM.XLF$time), KM.XLF$lower,
```

Parametric modeling of censored data

Likelihoods can be constructed for censored and truncated samples:

$$L \propto \prod_{det} f(x_i) \prod_{cens} (1 - S(x_i)) \prod_{trunc} f(x_i) / S(x_{trunc})$$

In some cases, the likelihood can be written in closed form.

With the likelihood, the full capabilities of MLE and Bayesian inference are available: parameter estimation with confidence intervals; model selection with penalized likelihoods; marginalization over uninteresting variables; etc.

This approach is not often used in astronomy.

Two-sample tests, correlation & regression

Several generalizations of 1930s nonparametric 2-sample tests (e.g. Wilcoxon) were developed in 1960-70s that treat censored values in reasonable fashions:

Gehan, logrank, Peto-Peto, Peto-Prentice tests

Generalized Kendall's tau correlation coefficients developed in 1970-90s.

Linear regressions developed during 1970-80s:

- Maximum likelihood line assuming Gaussian residuals (using EM Algorithm)
- **Buckley-James line** assuming nonparametric KM residuals
- **Cox regression** for multivariate independent variables
- Akritas-Thiel-Sen semi-parametric line

Gehan's survival 2-sample hypothesis test

What is the chance that two censored datasets do not arise from the same underlying distribution? $H_0 : S_1(x) = S_2(x)$

The Gehan test, developed in 1965 as a generalized Wilcoxon test for survival data, is perhaps the simplest of the censored two-sample tests. For a left-censored Sample 1 with n objects, $x_1^1, x_2^1, \dots, x_n^1$, and Sample 2 with m objects, $x_1^2, x_2^2, \dots, x_m^2$, compute the pairwise quantity

$$\begin{aligned} &= +1 \text{ if } x_i^1 < x_j^2 \text{ (where } x_i^1 \text{ may be censored)} \\ U_{ij} &= -1 \text{ if } x_i^1 > x_j^2 \text{ (where } x_j^2 \text{ may be censored)} \\ &= 0 \text{ if } x_i^1 = x_j^2 \text{ or if the relationship is ill-determined.} \end{aligned} \tag{10.20}$$

Gehan's test statistic is

$$W_{Gehan} = \sum_{i=1}^n \sum_{j=1}^m U_{ij}. \tag{10.21}$$

W_{Gehan} is asymptotically normal with zero mean. A common estimate of its variance of based on permutations is

$$\widehat{Var}(W_{Gehan}) = \frac{mn \sum_{i=1}^{n+m} U_i^2}{(n+m)(n+m-1)} \tag{10.22}$$

Bivariate correlation for censored data

Consider a nonparametric hypothesis test for correlation. Helsel proposes a generalizing Kendall's coefficient based on pairwise comparison of data points, (x_i, y_i) - (x_j, y_j) .

$$\tau_H = \frac{n_c - n_d}{\sqrt{\left(\frac{n(n-1)}{2} - n_{t,x}\right) \left(\frac{n(n-1)}{2} - n_{t,y}\right)}}$$

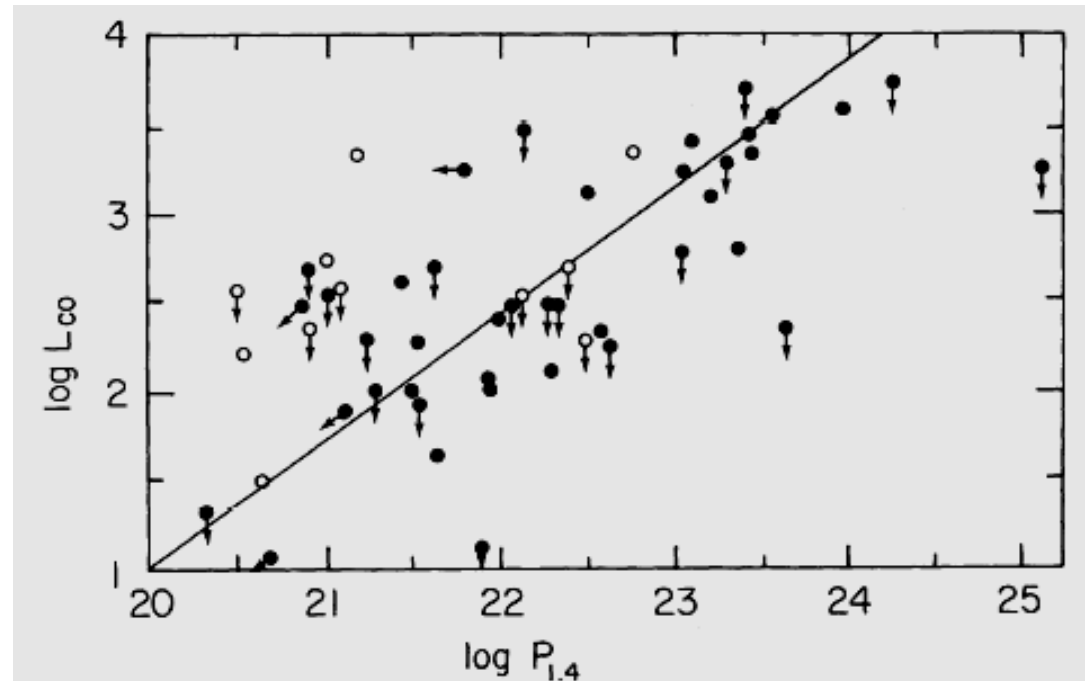
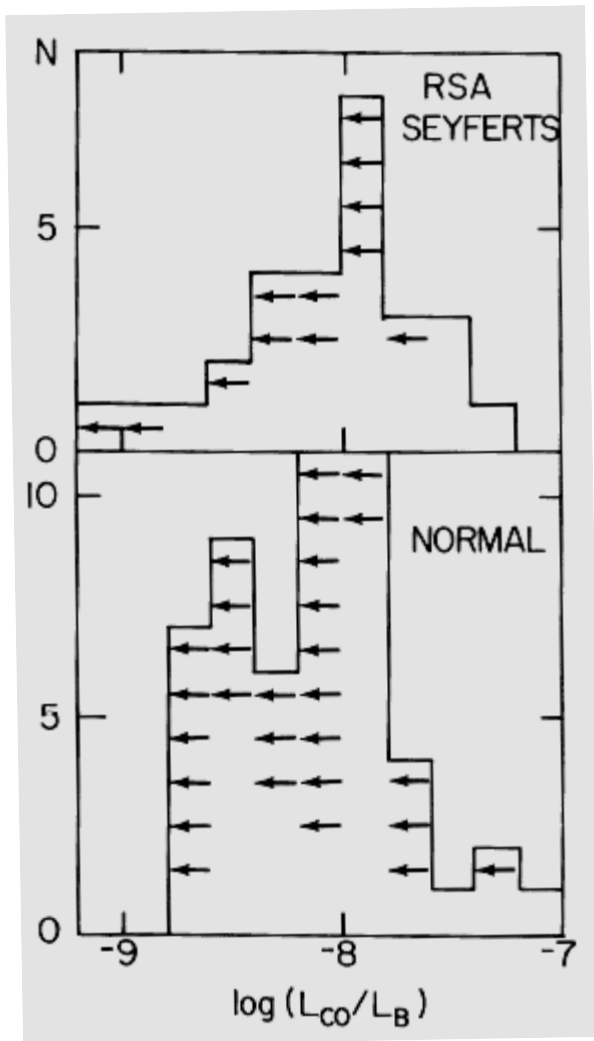
where n_c is the number of concordant pairs, n_d is the number of discordant pairs, $n_{t,x}$ and $n_{t,y}$ are the number of ties in x and y respectively.

As the censoring fraction increases, fewer points contribute to the numerator of τ_H , but the denominator measuring the number of effective pairs in the sample also decreases. So τ_H depends on the detailed locations of the censored points.

Linear regression: Several approaches

- ① Iterative least squares, the familiar linear regression model $y = \alpha + \beta\mathbf{x} + \epsilon$ where $\epsilon = N(0, \sigma^2)$. (Industrial reliability)
- ② Accelerated failure-time model $\log y = \alpha + \beta\mathbf{x} + \epsilon$. (Industrial reliability)
- ③ Tobit regression. (Econometrics)
- ④ Proportional hazards model (Cox regression) where the hazard rate has an exponential dependence on the covariates, $h(y|\mathbf{x}) = h_0(y)e^{\beta\mathbf{x}}$. MLE estimation and inference. (Biometrics)
- ⑤ Buckley-James line permitting non-Gaussian residuals around the line. ϵ is estimated in local regions of x using the Kaplan-Meier estimator. (Biometrics)
- ⑥ Akritas-Thiel-Sen line for doubly censored data shown in figure above. (Astronomy)

An example of astronomical censored data

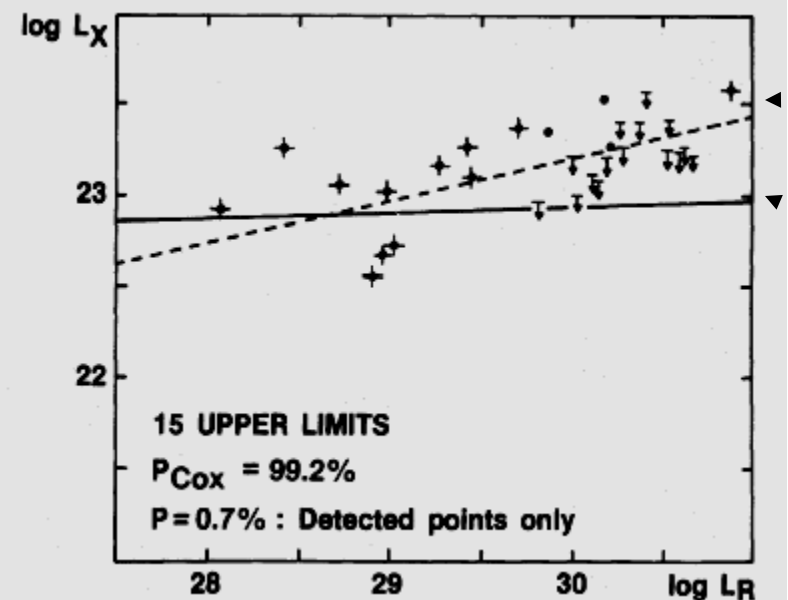
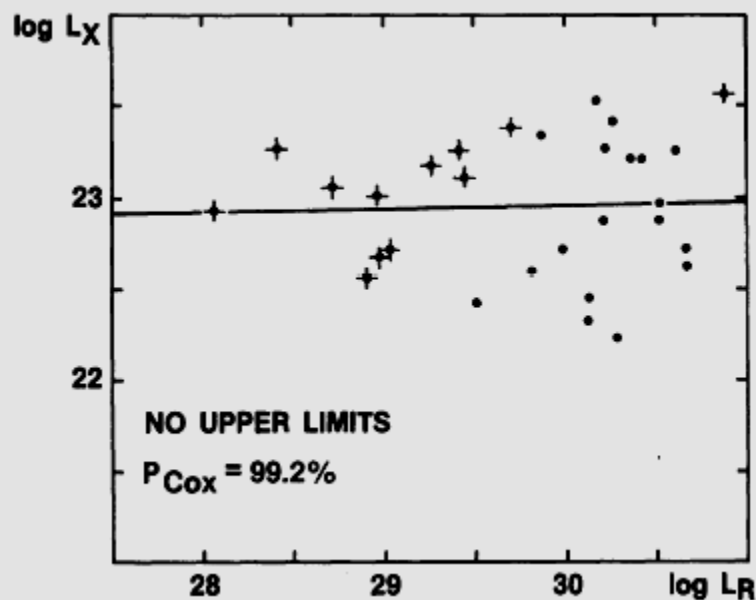


Heckman et al. 1989 A millimeter-wave survey of CO emission in Seyfert galaxies

Test of bivariate correlation/regression for simulated flux-limited surveys

Here is a simulation of uncorrelated X-ray and radio luminosities of a hypothetical sample of galaxies. The left panel has infinite sensitivity in the X-ray band, while the right panel shows finite sensitivity giving upper limits. A spurious correlation is found if only detections are considered (dashed line), but no correlation is found if survival methods (e.g., Cox regression) are used.

Biased line using detections only



Isobe, Feigelson & Nelson 1986

**Unbiased Buckley-James line
Including nondetections**

Software for survival calculations

Our group produced a stand-alone Fortran 77 code, Astronomy SURVival analysis (ASURV), which has been widely used. Most of its functionalities are in R/CRAN, and we recommend future work be conducted within R. ASURV implements:

Univariate distribution function Kaplan-Meier estimator with confidence limits & quantiles. Assumes random censoring.

Univariate two-sample tests Gehan, logrank, and Peto-Prentice tests. Can treat unusual censoring patterns.

Bivariate correlation coefficient Generalized Kendall's τ . Can treat censoring in both variables.

Bivariate linear regression MLE assuming normal residuals (EM Algorithm), Buckley-James line treating non-normal residuals, Schmitt's binned regression.

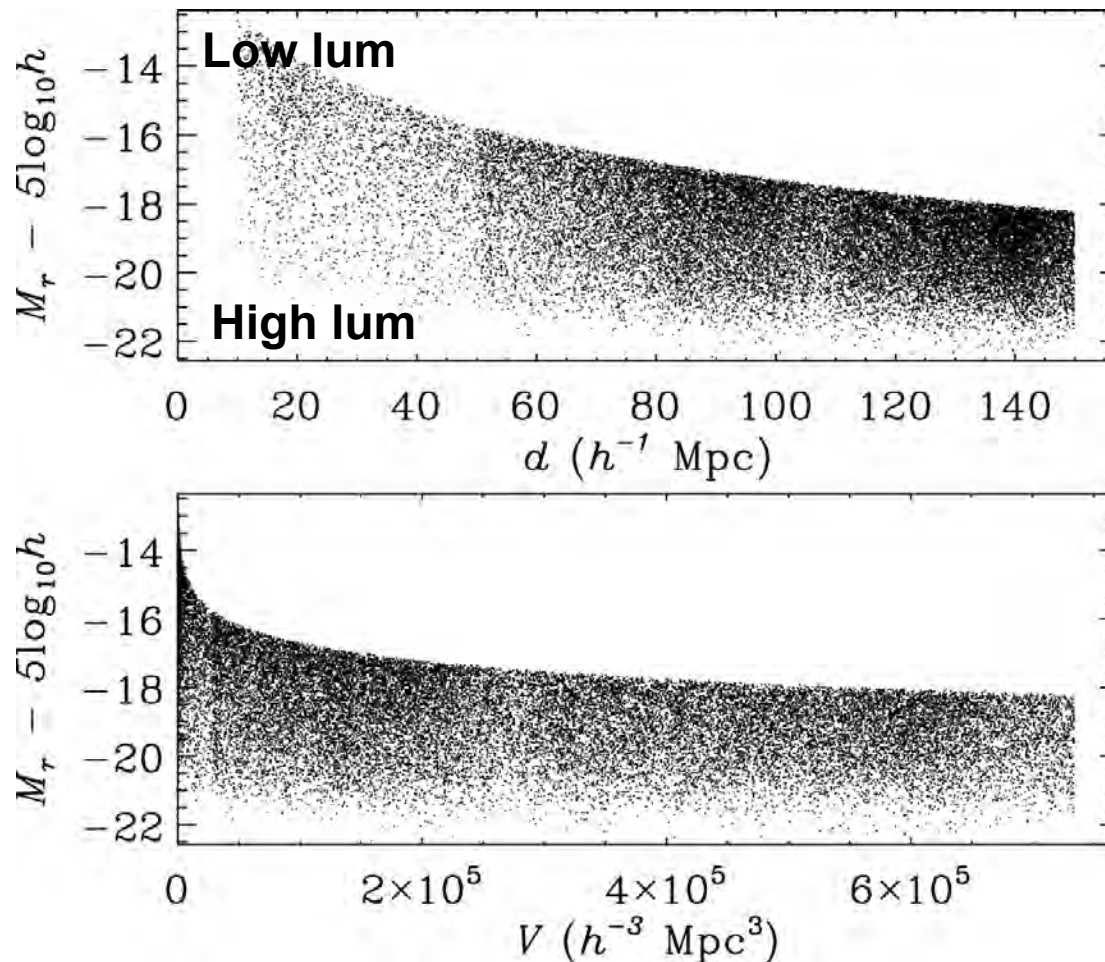
***But we now encourage use of R/CRAN
rather than the old ASURV !!***

Truncation in extragalactic surveys: The simple case of nearby galaxies



A recent SDSS study

Volume-limited survey of 28,089 with $d < 150$ Mpc



Blanton et al. 2005 The Properties and Luminosity Function of Extremely Low Luminosity Galaxies

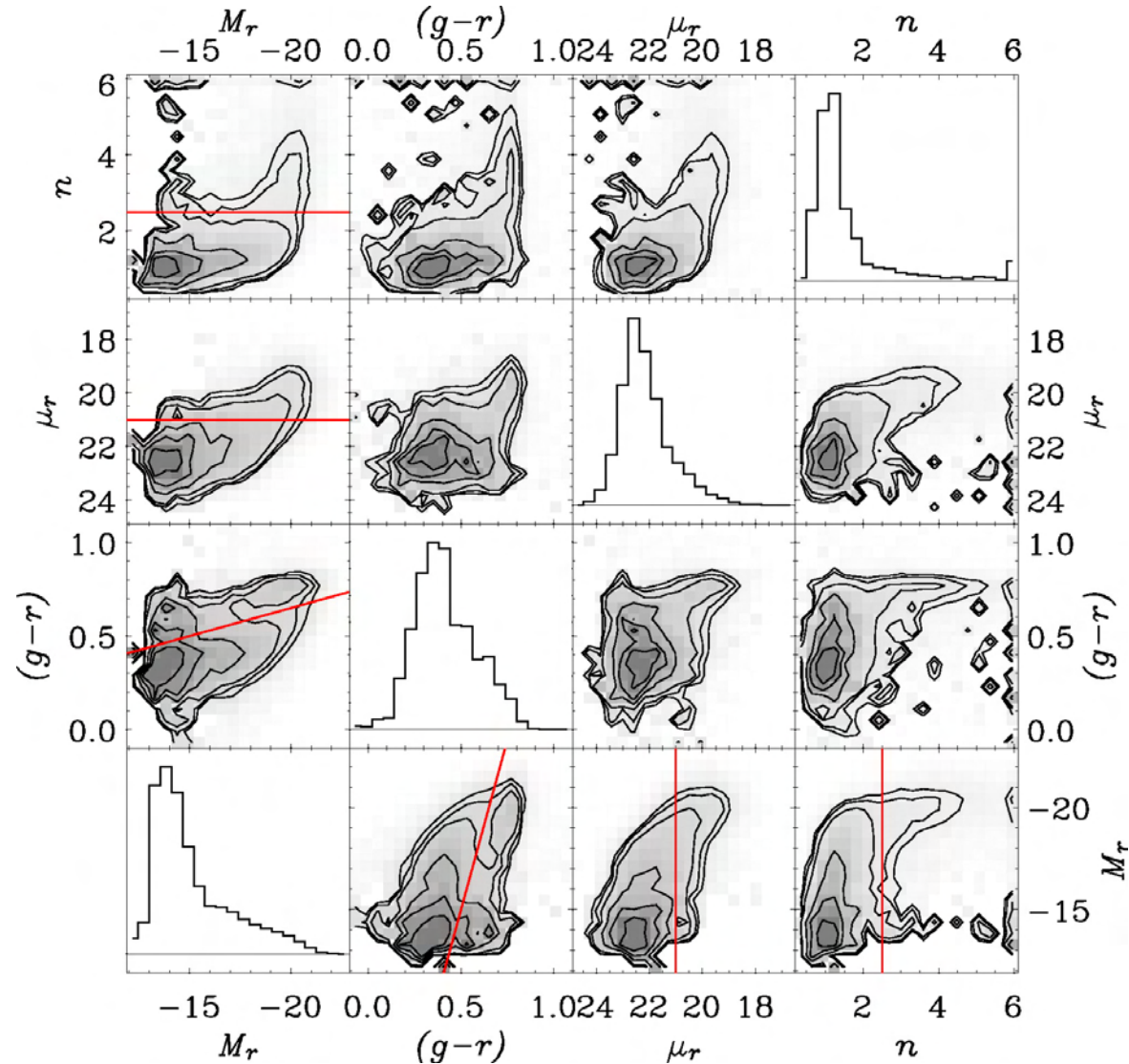
Ancillary properties are also measured

Concentration

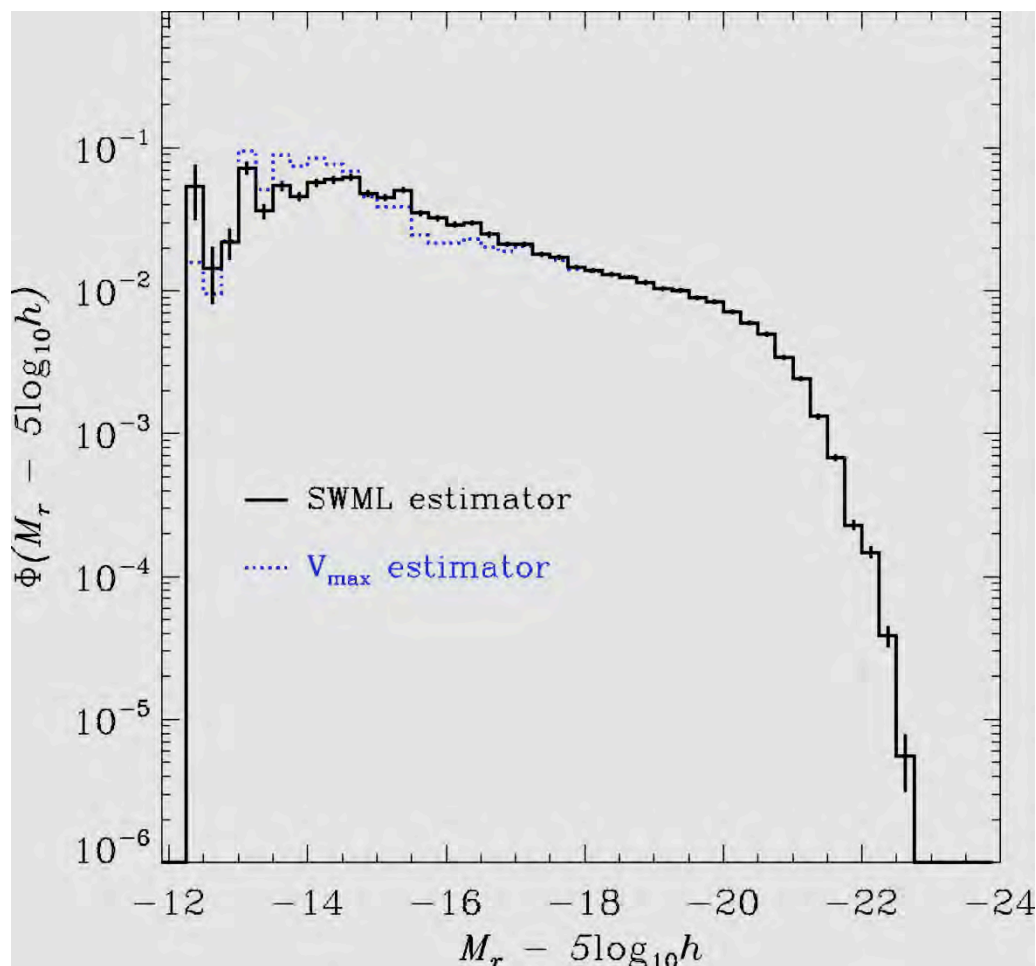
Surface brightness

Color

Luminosity



Normal galaxy luminosity function



**Parametric model:
Schechter function
= Gen. Pareto distribution**

$$dN/dL \sim \Phi \sim L^{-\alpha} e^{-L/L^*}$$

**Nonparametric methods:
two used here ($1/V_{\max}$,
stepwise max. likelihood)**

(Many technical issues concerning correction for missing low-surface brightness galaxies, K-correction and evolution-corrections to luminosity, double Schechter fits, etc.)

Nonparametric estimators to LFs

1. Classic estimator:

$$\Phi(L) = N(L) / V$$

where N is the number of stars/galaxies/AGNs in surveyed volume V . In flux-limited surveys, this is a (badly) biased estimator.

2. Schmidt's ' $1/V_{\max}$ ' estimator (>700 citations):

$$\Phi(L) = \sum 1 / V_{\max}(L_i)$$

where $V_{\max}$ is the maximum volume within which an object of the observed flux could have been seen given the survey's sensitivity limit. This can be an unbiased estimator but with high variance.

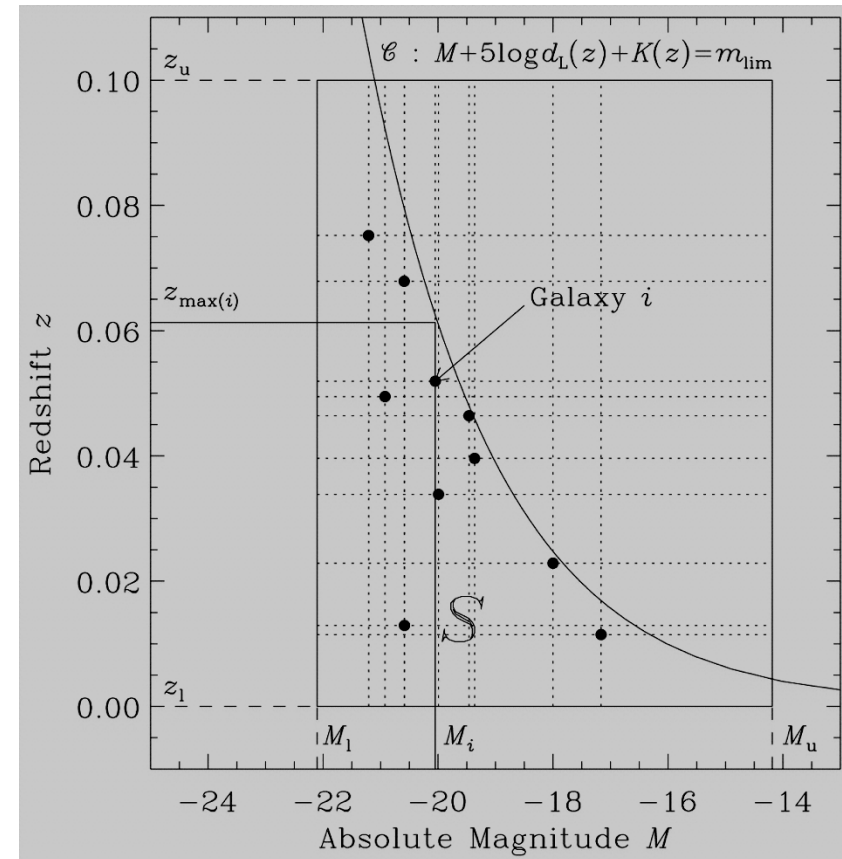
Schmidt 1968 Space Distribution and Luminosity Functions of Quasi-Stellar Radio Sources
Felten 1976 On Schmidt's V_m estimator and other estimators of luminosity function

Lynden-Bell-Woodroffe MLE (>100 citations):

Recursion relation:

$$\Phi(M) \propto \sum_{k=1}^{M_k < M} \psi_k = \psi_1 \prod_{k=1}^{M_k < M} \frac{C_k + 1}{C_k}.$$

where C_k is the number of stars/galaxies/AGNs in the k -th rectangle in the luminosity-distance diagram



Lynden-Bell 1971 A method of allowing for known observational selection in small samples applied to 3CR quasars

Woodroffe 1985 Estimation of a distribution function with truncated data

Stepwise maximum-likelihood estimator (>500 citations):

$$\ln \mathcal{L} = \sum_{i=1}^{N_{\text{obs}}} \left\{ \sum_{\ell=1}^K W(M_{\ell} - M_i) \ln \phi_{\ell} - \ln \left[\sum_{\ell=1}^K \phi_{\ell} H(M_{\text{lim}}(z_i) - M_{\ell}) \Delta M \right] \right\}. \quad (11)$$

where $W(M_{\ell} - M) \equiv \begin{cases} 1 & \text{for } M_{\ell} - \frac{\Delta M}{2} \leq M \leq M_{\ell} + \frac{\Delta M}{2}, \\ 0 & \text{otherwise.} \end{cases}$ (8)

$$H(M_{\text{lim}}(z_i) - M) \equiv \begin{cases} 1 & M_{\text{lim}}(z_i) - \frac{\Delta M}{2} > M, \\ \frac{M_{\text{lim}}(z_i) - M}{\Delta M} + \frac{1}{2} & M_{\text{lim}}(z_i) - \frac{\Delta M}{2} \leq M < M_{\text{lim}}(z_i) + \frac{\Delta M}{2}, \\ 0 & M_{\text{lim}}(z_i) + \frac{\Delta M}{2} \leq M, \end{cases}$$
 (10)

Set $d\mathcal{L}/d\phi = 0$ to obtain

$$\phi_k \Delta M = \sum_{i=1}^{N_{\text{obs}}} W(M_k - M_i) \times \left[\sum_{i=1}^{N_{\text{obs}}} \frac{H(M_{\text{lim}}(z_i) - M_k)}{\sum_{\ell=1}^K \phi_{\ell} H(M_{\text{lim}}(z_i) - M_{\ell}) \Delta M} \right]^{-1}. \quad (13)$$

Efstathiou et al. 1988 Analysis of a complete galaxy redshift survey.
II - The field-galaxy luminosity function

Takeuchi et al. (2000) apply these luminosity function estimators to simulations of small galaxy catalogs ($N=100$ and 1000). All perform well when spatial distribution is homogeneous. But for spatially clustered distributions, the $1/V_{\text{max}}$ estimator is badly affected.

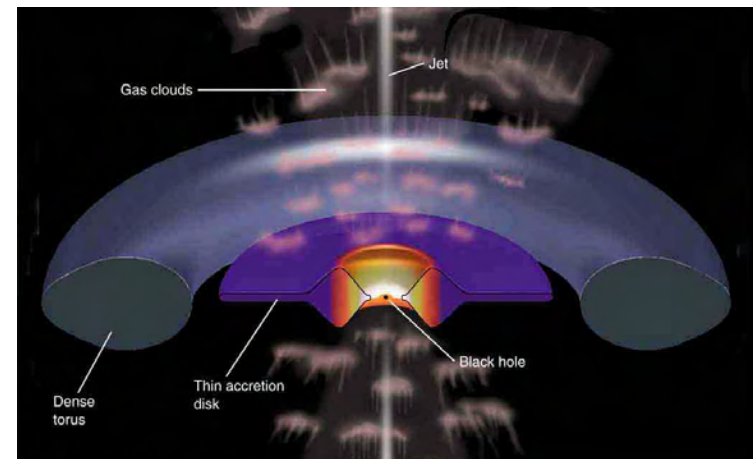
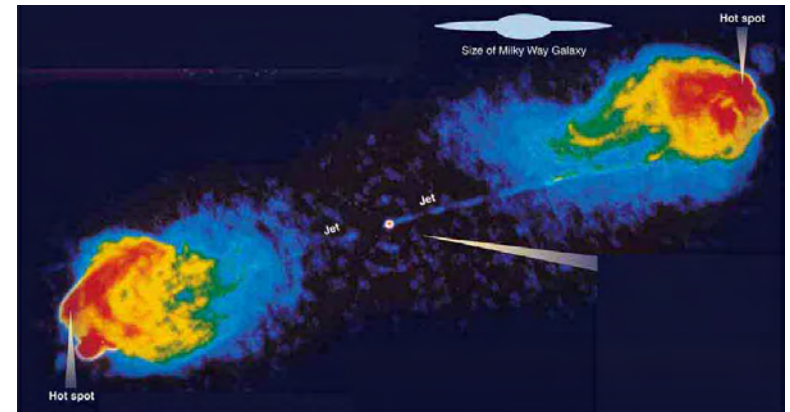
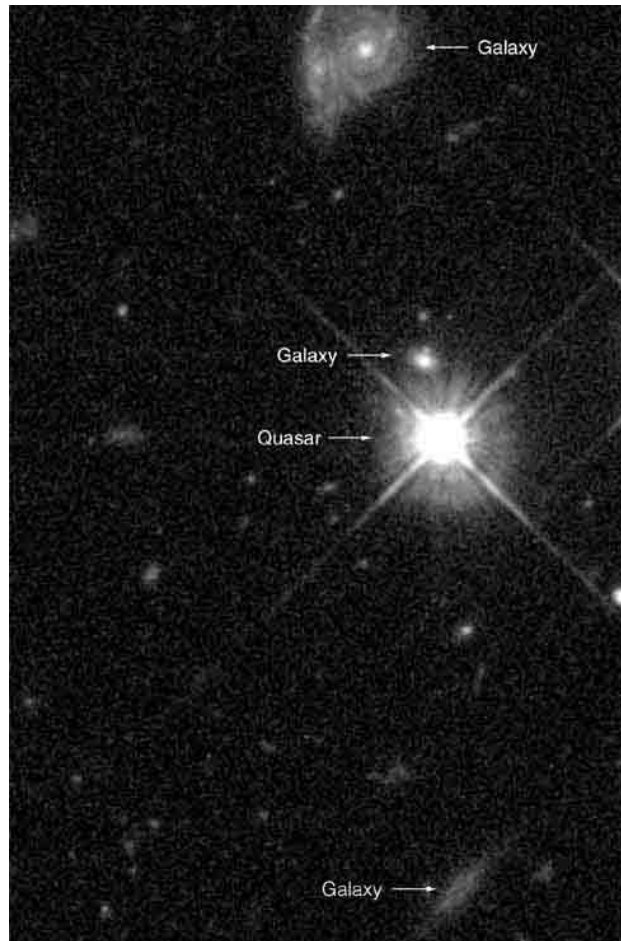
There has been no analytical evaluation of the mathematical properties of these estimators since study of $1/V_{\text{max}}$ by Felten (1976).

The Lynden-Bell-Woodroffe estimator for randomly truncated univariate data is mathematically the best choice: unbiased, unbinned, nonparametric maximum likelihood, asymptotically normal.

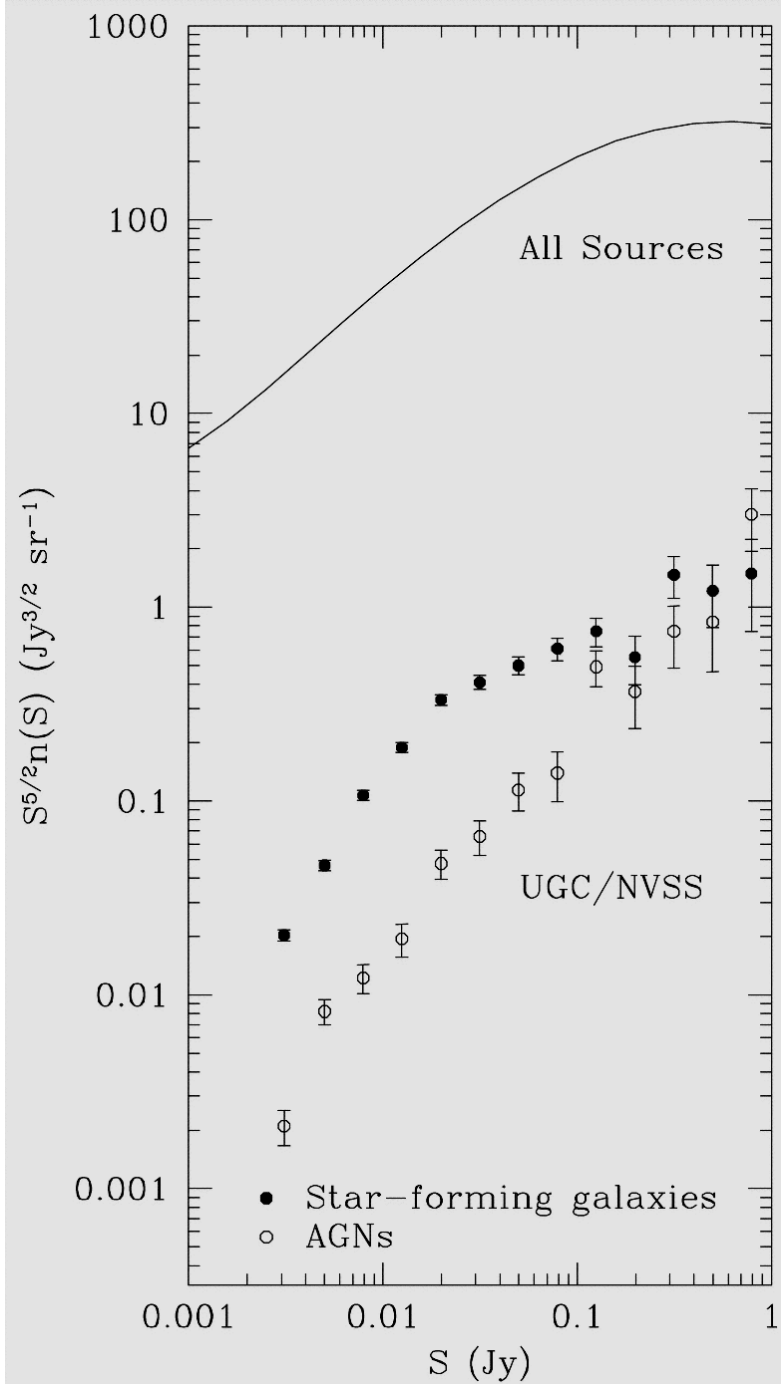
It is the analog of the Kaplan-Meier estimator for randomly censored univariate data.

Extragalactic surveys

The difficult case of quasar evolution



AGN exhibit huge cosmic evolution: hundreds of times more luminous quasars when the Universe was half its current age compared to today.



Here is the logN-logS distribution of radio sources in the sky. The expected $\log N \sim -1.5 \log S$ of a uniform distribution has been removed here, so deviations from a horizontal line indicates cosmic evolution of the volume density and/or LF.

The radio sources arise from two main classes, AGN and star-forming galaxies, which exhibit different evolution behaviors.

Enormous literature 1960-90s on quasar evolution, and a 1990-00s literature on star formation evolution (the 'Madau plot').

Condon et al. 2002 Radio sources and star formation in the local universe

Two or more flux limited surveys

1. A flux-limited survey gives a catalog of sources at one waveband
2. Astronomers observe this sample at another waveband, but only some objects are detected
3. Questions:
 - What is the LF in the second waveband? (density estimation)
 - Are these LFs the same in different subsamples? (k-sample test)
 - What is the relationship between the first and second luminosities? (correlation & regression)
4. Generalize to a multivariate database with many properties measured for the initial sample. Nondetections appear in many columns.

In astronomical parlance, the nondetections give upper limits.

In statistical parlance, the nondetections give left-censored data points.

These questions have arisen in hundreds of studies

Several solutions to censored LF problem were considered by astronomers during 1975-85, leading to the rediscovery of Efron's redistribute-to-the-right algorithm for the maximum-likelihood Kaplan-Meier product-limit estimator for randomly censored data.

Schmitt, Feigelson & Nelson and Isobe et al. introduced astronomers to existing methods of "survival analysis" for treatment of censored data.

Schmitt 1985 Statistical analysis of astronomical data containing upper bounds...

Feigelson & Nelson 1985 Statistical methods for astronomical data with upper limits

I - Univariate distributions

Isobe et al. 1986 Statistical methods for astronomical data with upper limits

II - Correlation and regression

ASURV code

Stand-alone code implementing survival methods for astronomy:

- **Kaplan-Meier estimator for the luminosity function**
- **Gehan, logrank, Peto-Peto & Peto-Prentice 2-sample tests**
- **Cox regression for bivariate correlation**
- **Generalized Kendall's τ & Spearman's ρ coefficient including censoring in both variables (Brown, Hollander & Korwar, Akritas)**
- **Linear regression with Gaussian residuals (EM Algorithm)**
- **Linear regression with K-M residuals (Buckley-James)**
- **Linear regression with censoring in both variables (Campbell, Schmitt)**

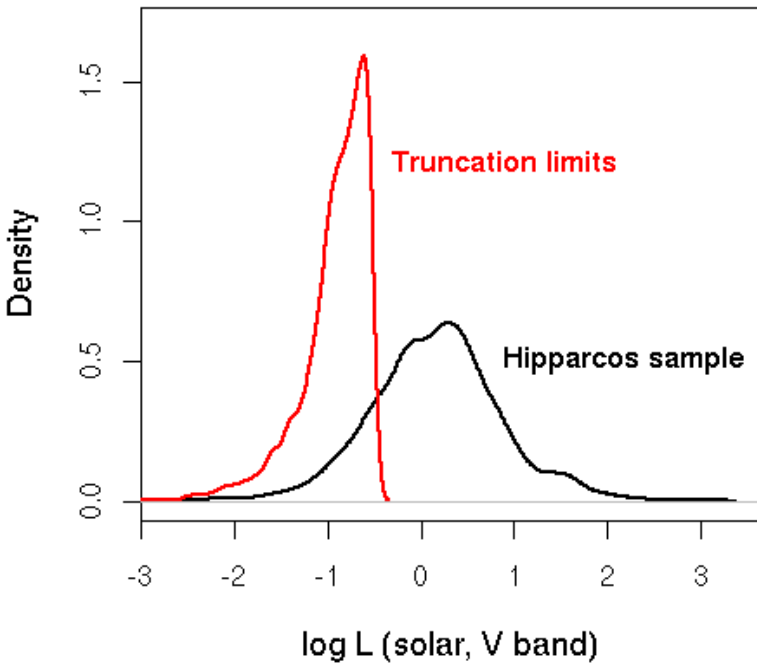
LaValley, Isobe & Feigelson 1992 ASURV software report

Survival analysis in R/CRAN

Base-R has six packages with functions for the Kaplan-Meier estimator (*survfit*), 2/k-sample tests (*survdifft*), and bivariate correlation (*survobrien*). Regression for univariate (*survfit*, *survreg*), bivariate (*bj*), & multivariate (*cph*, *coxreg*) problems, plus temporally dynamic models (*piecewise*, *dynreg*). Many ancillary functions (plotting, confidence intervals, smoothing, prediction, robustness, bootstrap, etc).

CRAN has ~180 packages with a bewildering variety of functions. Examples includes: confidence intervals (*km.ci*, *kmconfband*), constraints (*kmc*) and weighting (*MAMSE*) for the KM estimator, single/double truncation (*DTDA*), Cox regression (many), Bayesian models (many), multivariate classification (*rpart*, *party*, *kaps*), prediction error, simulation, and more. The *NADA* package is particularly useful for astronomy as it treats left-censored data. See summary at *CRAN Task View for Survival Analysis*.

Stellar luminosity function



Open circles: Known stellar LF from volume-limited sample

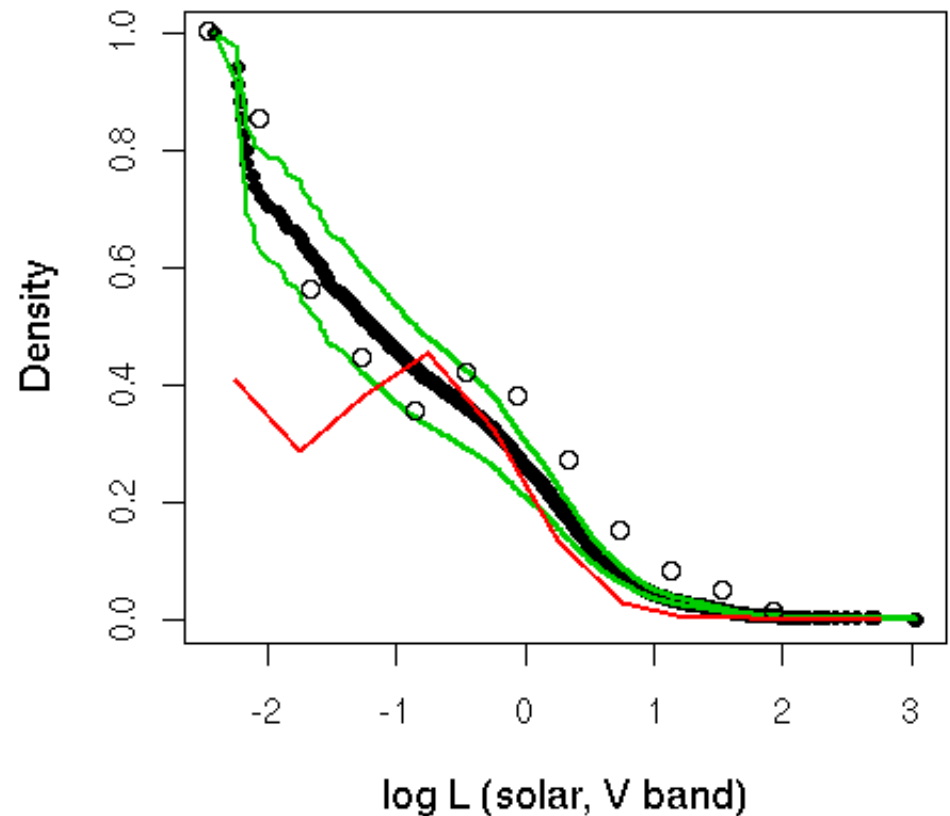
Black circles with green confidence bands: Lynden-Bell-Woodrooffe estimator

Red histogram: Schmidt's $1/V_{\max}$ estimator

computed with CRAN package DTDA

Data: $(L, L_{\min})$ for 3,307 Hipparcos stars with $V < 10.5$ and parallactic distance < 75 pc

Lynden-Bell-Woodrooffe estimator



Methodological work needed for astronomical nondetections

1. Investigate methods for astronomical censoring patterns

Type 1 censoring in $F=L/4pd^2$ gives nonrandom censoring in L

2. Develop multivariate analysis for censoring in all variables

Clustering, regression/PCA, MANOVA, MARS & neural nets, etc.

3. Calculate nonlinear regression & goodness-of-fit

4. Treat simultaneous censoring & measurement error

Unlike biometrical and industrial reliability testing environment where survival times are measured precisely, in astronomy the censored value is typically set at the 3σ upper limit where σ is the known noise level. A new statistical approach is needed that treats all observations (detected or not) with measurement errors. See B.C. Kelly (ApJ, 2007) for the beginning of a MLE approach.

5. Development of suite of tools based on the Lynden-Bell-Woodroffe estimator for truncated data

Two-sample tests, correlation coefficients (Efron-Petrosian), linear regressions, multivariate, ...

Censoring & truncation: A relationship with biometrics?

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Astronomical survey

Goal: Characterizing LFs and relationships in stars/gals/AGNs

Full population of stars/galaxies/AGNs

Truncated sample from flux-limited survey

Censored measurements of other properties

AIDS epidemic

Goals: Characterizing infected populations & mortality from AIDS

Full population of infected individuals

Truncated sample from symptomatic individuals

Censored measurements of survival

Treating simultaneous censoring & measurement error

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Conclusion

Many statistical issues arise in the scientific analysis of astronomical surveys, but censoring (upper limits, nondetections) and truncation (flux limits) are perhaps the most distinctive.

Extensive methods for censored data were developed for biomedical & other applications that can be effectively (though not always perfectly) applied to astronomical censoring. Fewer methods are available to overcome truncation where less information is available.

An introduction to this topic with R/CRAN examples appears in Chpt. 10 of our text *Modern Statistical Methods for Astronomy with R Applications* (Feigelson & Babu, Cambridge Univ Press 2012)